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Paleomagnetic study of some Cretaceous and Tertiary sedimentary rocks of the Klamath Mountains province, California

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ABSTRACT

Paleomagnetic investigation of Cretaceous outliers and Tertiary sedimentary strata of the Klamath Mountains province, and of onlapping Cretaceous strata, has shown the rocks to be largely remagnetized. Samples studied are from the Upper Jurassic to Upper Cretaceous Great Valley sequence, Upper Cretaceous Hornbrook Formation, Eocene Montgomery Creek Formation, and Oligocene(?) Weaverville Formation. Cretaceous samples that survived the remagnetization have a mean remanence direction that is very close to the expected direction of the Cretaceous magnetic field at the locality of the Klamath Mountains. Data from both primary and remagnetized samples suggest the possibility of 11.5° ± 15.8° of post-Cretaceous clockwise rotation of the Klamath Mountains province. None of the data from either the primary or remagnetized samples shows evidence of the large amounts (~70°) of clockwise rotation that other workers have measured for the lower Tertiary of the Oregon Coast Range. Our data indicate that the Oregon Coast Range and Klamath Mountains province did not behave as a single rigid block during the early Tertiary. They also suggest that any post-Oligocene rotation of the Klamath Mountains province is less than the approximately 30° post-Oligocene rotation recently proposed for a combined Oregon Coast Range–Klamath Mountains–Cascade Range block.

INTRODUCTION

Many recently published paleomagnetic studies for rocks from the western Cordillera of North America show magnetic directions that are widely divergent from expected directions. Comparison of these discordant directions with data from rocks of the appropriate age from the North American craton indicates that many areas along the west coast of North America have been transported great distances or have undergone large clockwise rotations, or both, since their original formation. Many of these displacements are quite young and extend well into the Tertiary. A paleomagnetic study by Simpson and Cox (1977), which included earlier data from Cox (1957, 1959), showed that a large part of the Oregon Coast Range had undergone a clockwise tectonic rotation during the early Tertiary. Rocks of early Eocene age were found (Simpson and Cox, 1977) to have rotated about 70° east of their expected Eocene direction; all but about 30° of this rotation had occurred by 30 m.y. ago. Results of a study of some Oregon Coast Range intrusive rocks by Beck and Plumley (1980) agree well with the results of Simpson and Cox (1977), as do data from a study near the Oregon-Washington border by Cox and Magill (1977). Other recent studies suggest that the Cascade volcanic arc may have also formed part of this rotating block (Bates and others, 1981; Beck, 1980; Magill and Cox, 1980, 1981). Although early Tertiary tectonic rotations have also been measured in western Washington, the uniform large-scale rotations seen in the Oregon Coast Range apparently do not include rocks north of the Columbia River (Wells and Coe, 1980) or in Oregon east of the Cascade Range (Beck, 1980).

Simpson and Cox (1977) pointed out that the rotations of the Oregon Coast Range should include the Klamath Mountains as well, because marine strata of the middle Eocene Flournoy Formation of Baldwin (1974) and Tyee Formation seem to lap onto the crystalline rocks of the Klamath Mountains province. Also, Magill and Cox (1980, 1981) suggested that depositional contacts between units of the Cascade Range, the Oregon Coast Range, and the Klamath Mountains require both the Klamath Mountains and the Oregon Coast Range to have been involved in the same post–25-m.y.-ago rotation (27° ± 7°) that their data indicate for the Cascade Range. Because the Klamath Mountains are critical to interpretations of the tectonic history of western North America, our study was made to obtain some paleomagnetic data from this province.

Principal rock units of the Klamath Mountains range from early Paleozoic to Late Jurassic in age. Rocks young enough to test the possibility of early Tertiary rotation are scarce. Paleozoic and Upper Jurassic strata have been intruded by calc-alkalic plutonic rocks, virtually all of which are Jurassic. However, as with many intrusive rocks, little or no evidence exists to determine the possibility or amount of postmagnetization deformation. For this reason, none of the plutons was sampled for this study. Instead, we chose to study the small isolated patches of Cretaceous

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marine sedimentary strata that overlie the older basement rocks in the southern part of the province. These are erosional remnants of a once-continuous mantle of rocks of the Great Valley sequence deposited on the Klamath Mountains. Paleomagnetic samples were obtained from four of these patches (localities K1 through K4, Fig. 1). Onlapping Upper Cretaceous strata of the Hornbrook Formation were sampled at localities K5 and K6 near Yreka (Fig. 1). Results for paleomagnetic samples obtained from the onlapping Cretaceous Great Valley sequence at the south end of the province were previously reported (Mankinen, 1978). Nonmarine sedimentary rocks of the Eocene Montgomery Creek Formation (locality T1) and the Oligocene(?) Weaverville Formation (localities T2, T3, T4) also were sampled (Fig. 1).

METHODS

Paleomagnetic samples were collected by use of a portable core drill and were oriented with a magnetic compass. A total of 150 oriented cores was taken from the localities shown in Figure 1. Samples were measured using a cryogenic magnetometer. The average intensity of remanent magnetization for a site (before magnetic cleaning) was generally low, ranging from 7.1×10^{-7} to 5.4×10^{-5} emu/g. Representative samples from each locality were subjected to alternating field (AF), thermal, or chemical demagnetization experiments. Progressive AF demagnetization experiments were performed using a three-axis tumbler, and the samples were demagnetized twice at each increment of alternating field to eliminate the effects of rotational remanent magnetization, following the method of Hillhouse (1977). Thermal demagnetization experiments were performed in air using a magnetically shielded oven with ambient fields of less than 10 γ. Samples for chemical demagnetization were prepared using the multicoring method described by Henry (1979) and were progressively leached in 9*N* hydrochloric acid until no directional changes between treatments occurred (up to 8 h total). At each step, the samples were rinsed and allowed to dry in ambient fields of less than 100 γ. After leaching, the samples were progressively AF demagnetized in the normal manner.

No single demagnetization procedure was found to be applicable for all samples, and so a stable end point had to be determined for each sample individually. For this reason, only enough samples were treated by any one of the cleaning methods described to determine a characteristic

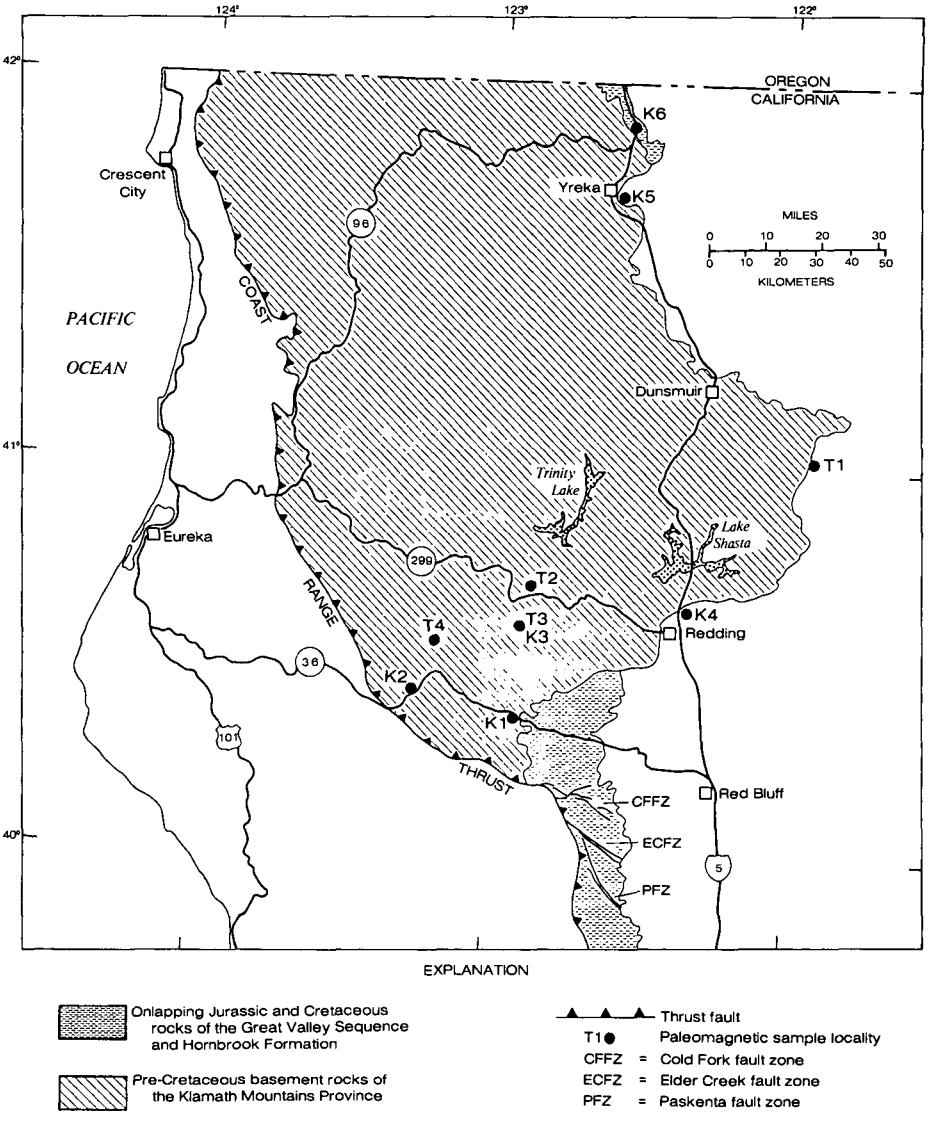


Figure 1. Map of Klamath Mountains province in northern California showing location of paleomagnetic sampling sites listed in Appendix 1.

TABLE 1. PALEOMAGNETIC RESULTS FOR ROCKS FROM KLAMATH MOUNTAINS PROVINCE

Locality	N	Uncorrected for bedding				Corrected for bedding			
		Incl.	Decl.	K	α_{95}	Incl.	Decl.	K	α_{95}
T4	5	31.3	343.8	64	9.7	16.1	338.0	64	9.7
T3		Indeterminate				Indeterminate			
T2	10	47.6	029.8	65	6.0	50.4	016.2	65	6.0
T1	10	57.8	001.3	52	6.8	57.8	001.3	52	6.8
K6	10	62.9	006.2	113	4.6	48.3	023.0	123	4.4
K5	7	65.9	008.3	21	13.4	61.5	019.4	21	13.4
K4	11	66.5	005.1	39	7.4	66.5	005.1	39	7.4
K3	17	61.4	011.9	37	6.0	85.6	064.9	32	6.4
K2	8	59.4	015.1	25	11.4	45.2	031.7	19	13.1
K1	13	63.5	001.2	25	8.4	66.2	058.0	23	8.9

Note: N = number of samples magnetically cleaned; Incl. = inclination, positive downward; Decl. = declination, east of north; K = precision parameter (N-1/N-R); α_{95} = 95% confidence oval about mean direction.

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direction for each locality. The mean remanence direction determined for each site is given in Table 1.

Strong-field thermomagnetic measurements were made on seven whole-rock samples from the different localities. These measurements were made in air using applied fields between 2,500 and 7,000 Oe at heating and cooling rates of 10 °C/min. The only magnetic mineral detected in these experiments was magnetite or Ti-poor titanomagnetite.

CRETACEOUS OUTLIERS

The Cretaceous sedimentary rocks at sites K1 through K6 (Table 1) have clearly been remagnetized after folding (Fig. 2). AF demagnetization was useful for isolating the remagnetization direction for these rocks, but thermal demagnetization was less effective. Some samples proved to be more easily cleaned with AF demagnetization after being treated with HCl. Magnetization directions of HCl-treated samples from localities K1, K3, K4, and K6 moved much closer to the site mean directions given in Table 1 than did the adjacent specimens that were cleaned by AF or thermal demagnetization alone. However, as previously mentioned, the directions in Table 1 no longer represent the primary remanence. These experiments show that some secondary components of magnetization in these rocks can be chemically removed but that the original remanence of the detrital magnetite has apparently been reset. With rare exceptions, none of the techniques could isolate the primary remanence in these rocks. The cause of the remagnetization is not known, but the grouping of the directions uncorrected for bedding attitude (Fig. 2) suggests that the remagnetization may be fairly recent. A few samples from these localities did retain a record of their primary remanence direction; these will be discussed below.

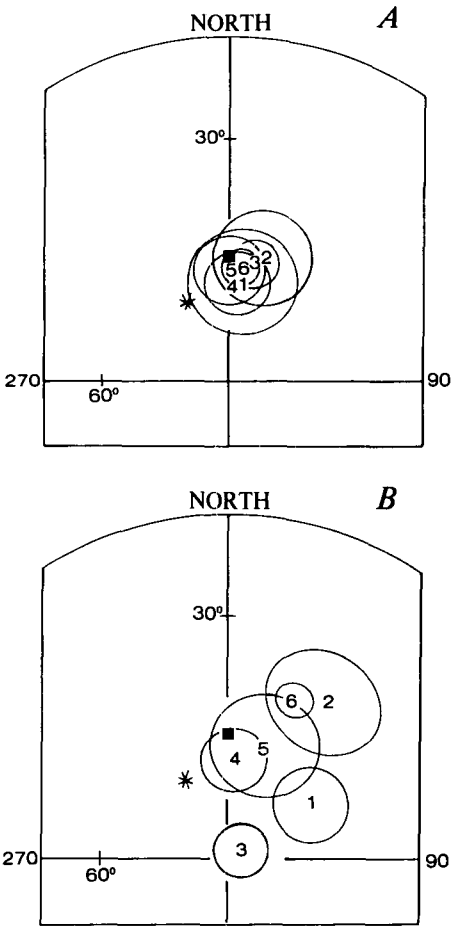


Figure 2. Average remanent magnetization directions after magnetic cleaning for sites within Cretaceous outliers. Sites before, A, and after, B, correction for attitude of bedding. Ovals, 95% confidence limits about average direction. Asterisk, expected Cretaceous field; square, average dipole field. Numbers 1 through 6 refer to sample localities K1 through K6 in Figure 1.

TERTIARY SEDIMENTARY ROCKS

Only four localities of Tertiary sedimentary rock visited during the course of this study proved to be suitable for paleomagnetic sampling. One of these, a thick (about 2 m), white tuff layer sampled along Reading Creek (T3 in Fig. 1) was found to have an unstable magnetization. Results from the other three localities are listed in Table 1. With the very small amount of data obtained, many more units must be sampled before meaningful conclusions can be reached on the basis of Tertiary sedimentary rocks alone.

GREAT VALLEY SEQUENCE

Any rotation of the Klamath Mountains province since the Cretaceous may have been accommodated by the Cold Fork, Elder Creek, and Paskenta fault zones (Fig. 1). The faults of these zones were active during the Cretaceous (Jones and others, 1969; Jones and Irwin, 1971; Bailey and Jones, 1973; Blake and Jones, 1977), and their cumulative displacement amounts to many tens of kilometres. They offset an Early Cretaceous (Valanginian) shoreline a minimum of 100 km, and the left-lateral sense of displacement may account for the westward bulge of the Klamath arc (Jones and Irwin, 1971).

A paleomagnetic study of the principal outcrop belt of the Great Valley sequence (Fig. 3), south of the Klamath Mountains, showed that those rocks are remagnetized and that the remagnetization probably occurred during Cretaceous time (Mankinen, 1978). If that interpretation is correct, and if there has been some early Tertiary rotation of the Klamath Mountains, then localities north and south of the fault zones should show some differences in their magnetization directions. Because no attempt was made to differentiate between these two areas in the earlier study, additional samples were magnetically cleaned to test this possibility.

The Dry Creek–Budden Canyon and South Fork Cottonwood Creek sections shown in Mankinen (1978) are north of the three fault zones, and the McCarty Creek, Grindstone Creek, and Putah Creek sec-

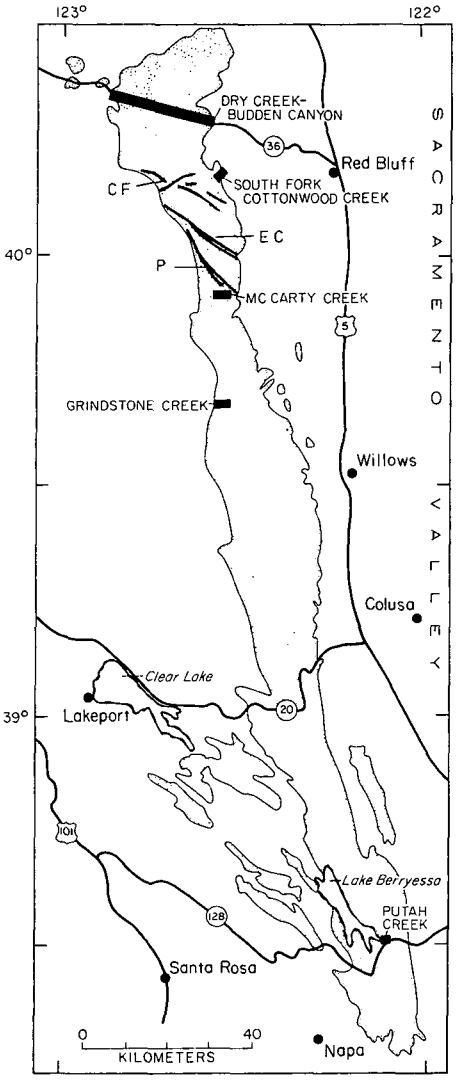


Figure 3. Locations and informal names of paleomagnetic sections within principal outcrop belt of Great Valley sequence (stippled)(after Mankinen, 1978). Offset of Cretaceous shoreline was left lateral along Cold Fork (CF), Elder Creek (EC), and Paskenta (P) faults.

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tions are to the south. Results after this additional magnetic cleaning are given in Table 2. Groups labeled Beegum Creek, Budden Canyon, and Dry Creek are all part of the Dry Creek–Budden Canyon section. Beegum Creek is at the western limit of this section, and these samples were collected from along its banks. Budden Canyon samples were collected along Highway 36 from Beegum Creek through Budden Canyon. The remaining samples were collected from the stream bed of Dry Creek. Because many samples from the Putah Creek section were shown to be unstable and generally unreliable (Mankinen, 1978), no additional measurements were made on rocks from this section. However, the data obtained from this study are included in Table 2 and compared with the other localities in Figure 4.

Inspection of Figure 4A shows that two of the southern localities have directions

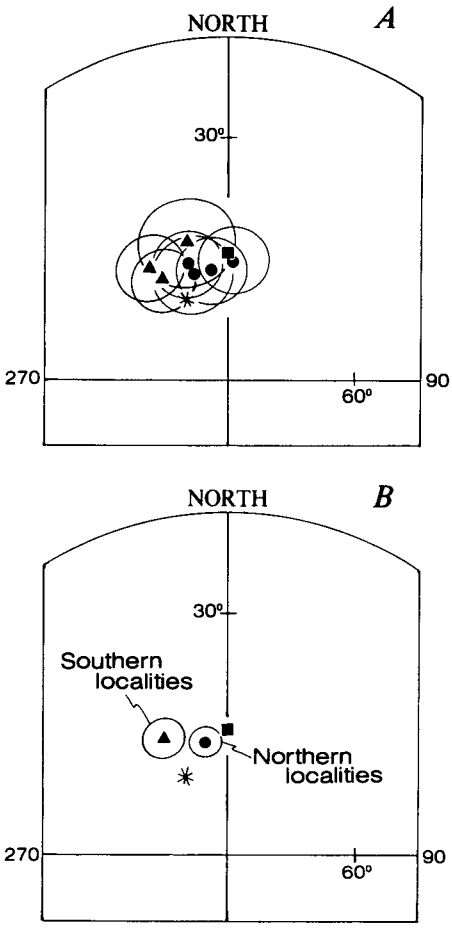


Figure 4. Directions of remagnetization for sites within Great Valley sequence. A, individual site mean directions. B, means of combined sites with associated 95% confidence limits. Localities north (dots) and south (triangles) of major fault zones shown in Figure 1. Other symbols as in Figure 2.

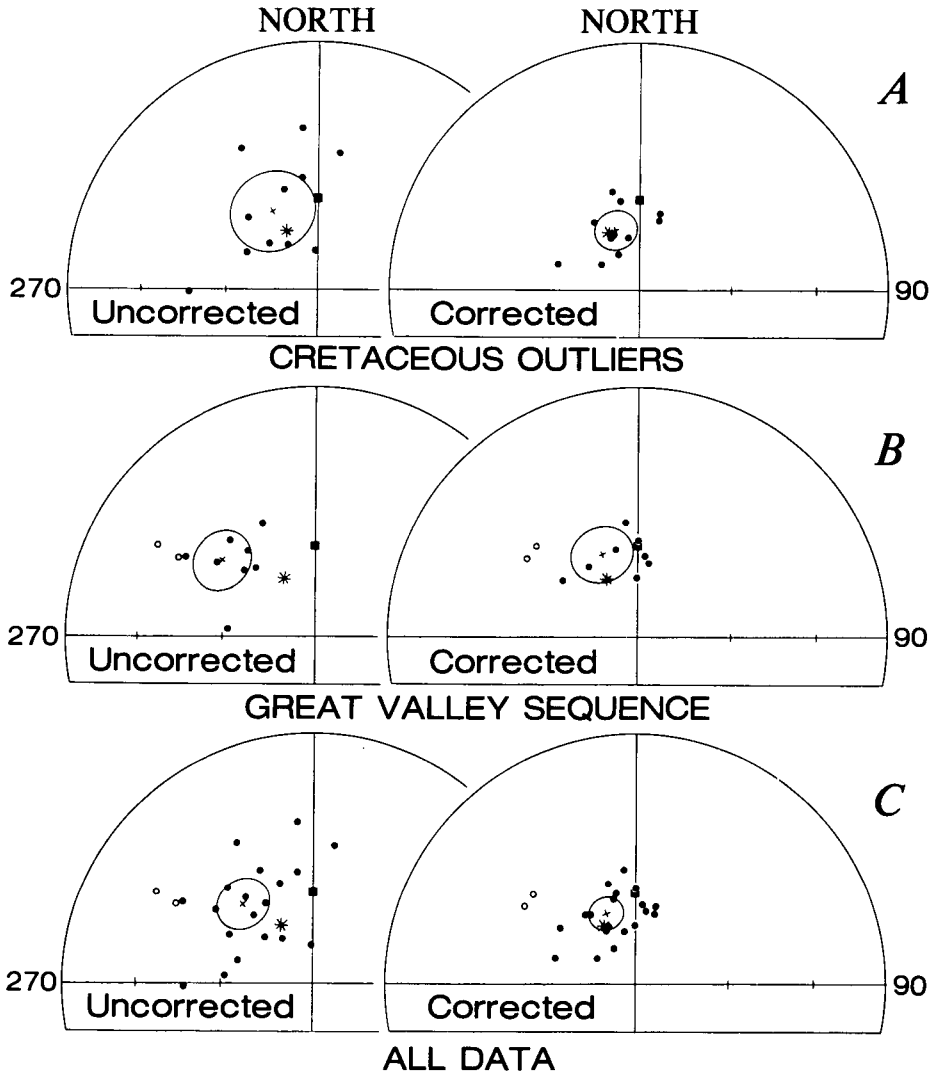


Figure 5. Mean remanent magnetization directions (x) and 95% confidence limits (ovals) for samples retaining their original magnetization. Samples from A, Cretaceous outliers; B, Great Valley sequence; and C, all data shown both uncorrected and corrected for attitude of bedding. Open circles are reversed directions projected through origin onto lower hemisphere. Other symbols as in Figure 2.

TABLE 2. REMAGNETIZED DIRECTIONS FOR GREAT VALLEY SEQUENCE

Locality	N	Incl.	Decl.	K	α_{95}
Beegum Creek	7	61.7	002.6	57	8.1
Budden Canyon	9	64.0	351.5	41	8.2
South Fork of Cottonwood Creek	7	63.9	342.0	40	9.6
Dry Creek	8	61.1	341.1	50	7.9
Average of northern localities	31	62.9	349.2	45	3.9
Grindstone Creek	9	57.4	324.5	43	7.9
McCarty Creek	8	61.6	326.2	56	7.5
Putah Creek	8	55.9	343.2	25	11.4
Average of southern localities	25	58.6	331.3	35	5.0

Note: See Table 1 for explanation.

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TABLE 3. RESULTS FROM UNREMAGNETIZED SAMPLES FROM GREAT VALLEY SEQUENCE

	N	Uncorrected for bedding				Corrected for bedding			
		Incl.	Decl.	K	α_{95}	Incl.	Decl.	K	α_{95}
Cretaceous outliers	11	59.7	329.9	13	13.2	68.7	388.7	45	6.9
Great Valley sequence	10	49.9	309.4	27	9.4	60.6	336.9	27	9.5
All data	21	55.3	318.7	16	8.3	64.9	337.7	33	5.6
Cretaceous, stable North America						68.5	332.0		

Notes: Inclination and declination for the Cretaceous is that which would be produced at the sampling localities by the Cretaceous reference pole given by Mankinen (1978). See Table 1 for further explanation.

that are distinctly westward of those of the northern localities. The third southern locality is Putah Creek, which has a large uncertainty about its mean direction and is not statistically distinguishable from the other localities. Combining the samples from all localities within each area into a northern and a southern group shows these differences quite well (Fig. 4B) and suggests that displacement of the areas has occurred since the rocks were remagnetized. The timing of this remagnetization should again be considered. Although the inclinations in Table 2 are somewhat shallower than previously determined (Mankinen, 1978), and shallower than expected for the Cretaceous field at this locality, the consistently westward declinations argue against the remagnetization being any younger than latest Cretaceous or earliest Tertiary. It appears from Figure 4 that left-lateral movement of the Klamath Mountains along the Cold Fork, Elder Creek, and Paskenta fault zones has resulted in a small amount of clockwise rotation of the province. The direction of the northern localities is 17.2° ± 8.6°east of the average Cretaceous field.

PRIMARY DIRECTIONS

A few samples, both from the Cretaceous outliers and the Great Valley sequence north of the major fault zones, seem to have retained their original directions of magnetization. These samples were clearly divergent from the main groups of samples defining the remagnetization directions shown in Figure 2 and Table 1 and were typically two standard deviations or more away from the calculated mean direction. For example, a comparison of the individual unremagnetized samples from the Cretaceous outliers plotted in Figure 5A (uncorrected) with Figure 2A shows that these samples are commonly to the west of, or shallower than, the mean remagnetization direction. Possibly unremagnetized samples were found at all sites except K4. The same situation holds for samples from the Great Valley sequence shown in Figure 5B. Mean directions before and after applying structural corrections are given in Table 3. Correcting samples from the Cretaceous outliers for the attitude of bedding yields a positive fold test with a marked decrease in the amount of scatter (Fig. 5A, corrected). Similar structural corrections within the Great Valley sequence produced no improvement in the statistical parameters, but the mean direction changed from one that did not reflect any known field direction to one that closely approximates the known Cretaceous field (Fig. 5B, corrected). Combining samples from both groups gives a mean that is indistinguishable from the Cretaceous field. The combination of a positive fold test and a direction of magnetization that agrees with the expected direction for the age of the rocks shows that the direction of these samples is probably primary. However, this finding does not rule out a small amount of post-Cretaceous rotation, because the uncertainty about this mean also encompasses the mean of the remagnetized Great Valley sequence samples north of the three fault zones. The clockwise rotation permissible by the unremagnetized samples is 5.7° ± 13.3°. Thus, when data from both the remagnetized and primary directions are compared, clockwise rotations greater than about 20° since the Cretaceous seem unlikely.

CONCLUSIONS

Material of an appropriate age to test the possibility of early Tertiary rotation of the Klamath Mountains province proved to be magnetically less than ideal. Remagnetization of these sedimentary rocks was almost total in most of the sections sampled. Although remagnetization can seldom be assigned a precise age, Earth's magnetic field during the Cretaceous and into the lower Tertiary was distinctive enough that a fair amount of confidence can be given to the postulated Late Cretaceous or early Tertiary age for the remagnetization of the Great Valley sequence. The steep magnetic inclination of the Cretaceous outliers also seems to preclude a time of remagnetization later than the early Tertiary. A small amount of clockwise rotation since that time could account for the north to northeast magnetic declinations of the remagnetized sites as seen in Figure 2. Although samples that survived remagnetization were rare, those that did survive record a reasonable Cretaceous direction. The limited Tertiary data are also entirely compatible with the hypothesis of little or no rotation. Averaging the rotations permitted by the primary and remagnetized samples yields a possible clockwise rotation since the Cretaceous of 11.5° ± 15.8° with 95% confidence, if rotation has indeed occurred. Any large amounts of early Tertiary rotation of units in the Oregon Coast Range must have occurred independently of the Klamath Mountains province.

It remains uncertain whether the possible small amount of post-Cretaceous rotation suggested by our study is related to post-Oligocene rotation of the western Cascade Range. However, if the Klamath Mountains province has rotated since the Cretaceous, the amount would seem to be far less than that required by the Magill and Cox (1980, 1981) model.

POSTSCRIPT

While this paper was in review, we were informed of recent paleomagnetic investigations in the western Cascade Range of northern California immediately adjoining our area to the east. Craig (1981) has measured an apparent 11.7° ± 10.9° clockwise

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rotation for the southern Cascade Range, based on a study of 44 lava flows that range in age from middle Oligocene to early Miocene. The nearly identical results from the two areas strongly suggest that the Klamath Mountains and southern Cascade Range may indeed have rotated as a single block during the middle Tertiary. Results of additional sampling in the Cascade Range (Craig and others, 1981) suggest further that the rotation may have occurred entirely during the Oligocene.

APPENDIX 1. SAMPLE LOCALITIES

All maps listed are U.S. Geological Survey 7.5-minute (scale 1:24,000) and 15-minute (scale 1:62,500) topographic quadrangle maps. Locality numbers are those used in text.

T4. Lat 40.55°N, long 236.82°E. Hayfork (1951) 15-minute quadrangle. Weaverville Formation. North bank of Hayfork Creek in Hayfork about 100 m east of bridge near west edge of sec. 12, T. 31 N., R. 12 W. Bedding strikes 210.0° E, dips 20.0° NW.

T3. Lat 40.59°N, long 237.11°E. Weaverville (1950) 15-minute quadrangle. Weaverville Formation. Both banks of Reading Creek about 1.9 km southeast of Clements Ranch. Center, sec. 33, T. 32 N., R. 9 W. Bedding attitude ranges from 005.0° E, dip 15.0° E, to 045.0° E, dip 12.0° SE.

T2. Lat 40.69°N, long 237.14°E. Weaverville (1950) 15-minute quadrangle. Weaverville Formation. West side of bridge crossing creek about 0.8 km southwest of Lowden Ranch. Near west edge of sec. 26, T. 33 N., R. 9 W. Bedding strikes 190.0° E, dips 12.0° W.

T1. Lat 41.02°N, long 238.09°E. Big Bend (1961) 15-minute quadrangle. Montgomery Creek Formation. North side of small tributary entering Pit River about 0.4 km north of bridge at Big Bend. Northeast corner of sec. 36, T. 37 N., R. 1 W. Bedding horizontal.

K6. Lat 41.88°N, long 237.44°E. Hornbrook (1955) 15-minute quadrangle. Hornbrook Formation. West side of northbound lanes of Highway I-5 (U.S. 99) beginning about 0.3 km north of BM 2092 and ending about 1.5 km southeast of Henley-Hornbrook freeway exit. East edge, sec. 32 and southeast corner, sec. 29, T. 47 N., R. 6 W. Bedding attitude ranges from 300.0° E, dip 15.0° NE, to 325.0° E, dip 20.0° NE.

K5. Lat 41.71°N, long 237.39°E. Yreka (1954) 15-minute quadrangle. Hornbrook Formation. About 2.2 km east of intersection of Oberlin Road and Highway I-5 (U.S. 99) immediately south of Yreka. Northeast corner, sec. 35, T. 45 N., R. 7 W. Bedding strikes 335.0° E, dips 07.0° NE.

K4. Lat 40.64°N, long 237.64°E. Project City (1957) 7.5-minute quadrangle. North side of bridge crossing Salt Creek about 2.4 km south of Pine Grove. Southeast corner, sec. 7, T. 32 N., R. 4 W. Bedding horizontal.

K3. Lat 40.60°N, long 237.09°E. Weaverville (1950) 15-minute quadrangle. First outcrop sampled on east side of creek where road crosses Browns Creek. Northeast corner, sec. 31, T. 32 N., R. 9 W. Bedding attitude ranges from 090.0° E, dip 23.0° S, to 95.0° E, dip 30.0° S. Second outcrop sampled about 0.1 km west of main road intersection at Clements Ranch. Northeast corner, sec. 29, T. 32 N., R. 9 W. Bedding strikes 095.0° E, dips 25.0° S.

K2. Lat 40.41°N, long 236.75°E. Pickett Peak (1954) 15-minute quadrangle. Two localities sampled on north side of Highway 36 about 0.3 and 0.7 km east of Glade Creek. Bedding at first locality ranges from 340.0° E, dip 14.0° NE, to 360.0° E, dip 20.0° E, and at second locality strikes 330.0° E, dips 15.0° NE. Third locality about 0.2 km from highway along banks of Glade Creek. Southeast corner, sec. 31, T. 30 N., R. 12 W. Bedding attitude ranges from 315.0° E, dip 30.0° NE, to 335.0° E, dip 20.0° NE.

K1. Lat 40.36°N, long 237.11°E. Chanchelulla Peak (1951) 15-minute quadrangle. Highway 36 about 0.4 km southwest of BM 2367 overlooking the settlement of Platina. Southwest corner, sec. 16, T. 29 N., R. 9 W. Bedding strikes 035.0° E, dips 25.0° SE.

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